



Recent technologies for transforming textile waste into value-added products: A review

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ABSTRACT

The disposal of textile waste has become a growing issue worldwide. The rising consumption of clothing and textile materials has resulted in high waste generation. This could adversely impact environmental health, including humans, animals and plants. The three textile recycling methods can be divided into mechanical, chemical, and biological processes. There has been a focus on mechanical and chemical processes, pyrolysis, enzymatic hydrolysis, biological recycling, and microbial engineering in the area of textile waste management. This review highlights the important parameters that affect the textile recycling performance and are significant for the success of the transformation process. To reach the zero-waste goal, textile recycling and converting this waste into value-added bioproducts are the necessary steps. The present review addresses the current status of textile recycling strategies, and the valorisation processes for converting textile waste materials into value-added products like biofuels, bioplastics, and others sustainable materials.

Introduction

Textile wastes are a major problem for the environment (Ütebay et al., 2020) when they are discarded inappropriately. The fashion industry has contributed significantly towards textile waste generation, through its fast fashion business strategy, characterized by large-scale production, variety, and affordability (Todor et al., 2022) and an expectation that the product will be discarded after a limited time-in-use. Introduction of sustainable practices is required in fashion industry to mitigate harmful environmental impacts of textile waste (Niinimäki et al., 2020). Recycling of textiles is, therefore, a potentially beneficial process as it could contribute significantly to the fashion industry becoming sustainable (Filho et al., 2019), if waste fabrics are used

in a circular economy fashion to make new textiles and non-textiles products (Okafor et al., 2021). There are many benefits to adopting a circular economy in the face of resource depletion and disposal issues (Wojnowska-Baryła et al., 2022).

Fabrics can be made from a variety of different materials. This leads to difficulty in the bioprocessing of textile waste because they contain natural and synthetic non-cellulosic fibres. Various strategies like mechanical, thermal, and chemical processes can be applied to recycle discarded textiles (Jeihanipour, 2011; Radhakrishnan, 2014; Pensupa et al., 2017; Baloyi et al., 2023). The cellulosic part of textiles can be bio-converted into purified value-added products (Subramanian et al., 2021). Cellulosic fibers are recyclable and have low density and cost. Recycling cellulosic fibers, especially chemically pulped fibers, can

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cause a loss of physical strength (Liu et al., 2019). Petrochemical-based synthetic fibers are not easily recycled because they are non-biodegradable and unsustainable. Conventional recycling methods based on mechanical and chemical processes are inadequate for the increasing amount of synthetic fiber waste. Petrochemical-based fibers have a high melting point, high crystallinity, stable molecular structure, excellent mechanical properties, and good hydrolytic and chemical corrosion resistance (Joseph et al., 2023). This means that they degrade very slowly in the natural environment, causing many ecological and environmental problems. In contrast, natural fibers such as wool, cotton, and silk are recyclable and biodegradable. Mixed fibers that contain both natural and synthetic fibers are difficult or costly to recycle (Suriani et al., 2021).

Because of the intricate composition of textile fibres, sorting is required before any of the waste can be recycled. A variety of fibres and colors used in fabrics makes sorting difficult, and reduces the quality of recycled materials. It is necessary to separate textiles from other waste prior to sorting them to be reused. There is a pressing need for an efficient strategy for identifying textile materials to boost the use of recyclable materials (Subramanian et al., 2020).

This review discusses the current status and novel technologies for effective recycling of textile wastes, and their valorisation into bio-products and other value added materials.

The global textile production and waste scenario

More than half of the 109 million tonnes of fibre produced in 2020 was polyester, according to a report by Textile Exchange, a group that promotes environmentally friendly textile manufacturing (Haji and Naebe, 2020). Polyester is the dominant fibre in the textile industry at global level (Fig. 1). Researchers reported that the microfibres released from textile waste fabrics during washing (Allary, 2021) are absorbed in aquatic life forms and can be harmful to humans if consumed. According to the United States Environmental Protection Agency (US EPA), textile waste takes up about 5 % of landfill space. On the other hand, only 15 % of textile waste is recycled, meaning that 85 % of it ends up in landfills (Grover et al., 2021).

These numbers and, therefore, the scale of the future problem are increasing at an incredible pace, as current textile recycling rates are less than 0.5 % by mass according to Tran and co-workers (Tran et al., 2022).

Environmental sustainability in textile recycling in switching to bio-based alternatives

The environmental impact of textile wastes can be reduced by recycling textiles when compared to the other options of incineration and landfilling (Sandin and Peters, 2018). Reuse, on the other hand, is especially beneficial compared even to recycling, and further improved when the reusing phase is prolonged and/or no or minimal additional works are done on the textile (i.e. they are reused for the function for which they were originally designed, in the form that they were originally produced). There are – thus a variety of ways to extend the usefulness of textile products to make them overall more sustainable.

Recycling involves the reprocessing of textiles (Riba et al., 2020). When a recycled product is better or more valuable than the original, the process is referred to as upcycling, while downcycling refers to the opposite (Ribul et al., 2021). Closing-the-loop on recycling refers to the reuse of materials from one product into a new product which is similar to the original. When material from one product is recycled and repurposed into another, the process is referred to as open-loop recycling.

The raw material used to make a polymer product is recovered and reprocessed in a closed-loop recycling system, creating a new polymer product of comparable quality to the original. Textile recycling involves primary to tertiary methods (Radhakrishnan, 2017; Forrest, 2019). Industrial waste and by-products are generally broken down mechanically during primary and secondary recycling. Textile waste can be turned into energy through pyrolysis and incineration in tertiary and quaternary recycling. Quaternary recycling (within this domain) is a process where – fibrous solid waste is burned and used for heat generation.

Textile waste reuse and recycling contribute to environmental stability (Fig. 2) and prior work has identified that to maximise the conservation of resources while minimising ecological effect, upcycling and closed-loop recycling can be used (Schmutz and Som, 2022). In a textile tailoring facility, the highest ecological footprint was found in the use of raw resources category, followed by energy consumption (Moazzem et al., 2021). Thus, the products derived from primary resources can be replaced with those made from recovered resources resulting in significant environmental benefits. In a study, Baydar et al. (2015) discussed about the life cycle assessment of cotton textile products and overall improvement in sustainability.

Routes to recycling of textile wastes

Mechanical or chemical recycling is the most common methods for

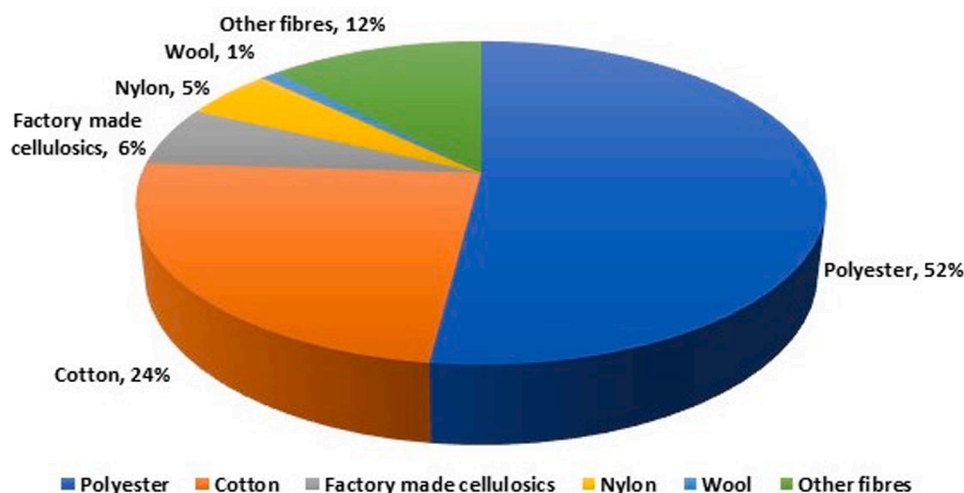


Fig. 1. Global fiber production share in 2020.

Source: CAS Content Collection, Ellen MacArthur Foundation, Nat. Rev. Earth Environ., Observatory of Economic Complexity, Textile exchange, (Bettenhausen et al., 2022)



Fig. 2. Strategies to reduce textile wastes from environment.

textile recycling. Using mechanical recycling, waste is transformed into useful materials for various applications, including construction, agriculture, and gardening (Briassoulis et al., 2013). In a review, Baloyi et al. (2023) discussed recent techniques for textile waste fabrics. They reported that the mechanical recycling method is limited in use because it degrades the quality of fabrics compared to chemical and biological methods. The depolymerisation or dissolution of polymers is an essential stage in the chemical recycling process (of cotton and viscose). It has been reported that fibres from chemical recycling can be of as high a quality as those from virgin materials (Valerio et al., 2020). It is possible to chemically treat sorted textile waste to extract resources such as cellulosic fibres for bioethanol production and proteins to generate wood panel adhesives (Rosenboom et al., 2022). In a recent study, Damayanti et al. (2021) discussed different possible mechanisms to

recycle textile waste materials (Fig. 3).

Pyrolysis

Pyrolysis can generate products in solid, liquid, and gaseous states from the decomposition of carbon polymers in waste textiles. Different textile materials that haven't been sorted prior to pyrolysis could be used in textile products (Limburg and Quicker, 2016). Without oxygen, pyrolysis transforms organic molecule waste into three-phase pyrolysates by reassigning C/H/O elements. Hydrocarbons and alcohols could be made from the syngas produced by thermochemical processes. Pyrolysis processes do not require pre-treatments, which makes them a viable option for treating polluted waste (Torri et al., 2021). Compared to chemical methods (including bio-chemicals), pyrolysis uses fewer chemicals, produces less waste, and can be scaled up more quickly. Chemical methods, on the other hand, require a greater number of chemicals, generate waste, and can require significant time for large scaleup of operations. Aromatic hydrocarbons are the most common liquid products made from textile waste (Yousef et al., 2019). In pyrolysis, the operating parameters influence the amount of oil and tar formed. At 500 °C, hydrocarbon compounds and aromatic oxygen were discovered. Furthermore, alkylphenols were produced at > 600 °C, and the O₂ yield began to decline at temperatures above 800 °C (Zhou et al., 2016). Aromatic compounds with no substituent groups, naphthalene or benzene, were found in the condensable compounds at 800 °C. In contrast, the aromatic hydrocarbon compounds with three and four rings were produced when the reaction temperature was around 850 °C (Ravenni et al., 2019).

A pyrolysis reactor converts cotton textile waste into carbon nanoball (CB) with a 10–20 nm diameter (Yousef et al., 2021). In concrete composite applications, textile waste pyrolysis char can be used as a primary filler. But Jagdale et al. (2017) investigated how cotton waste pyrolysis carbon could be used in active electrode battery materials. Cotton waste pyrolytic char can also be used as an adsorbent. When measuring the

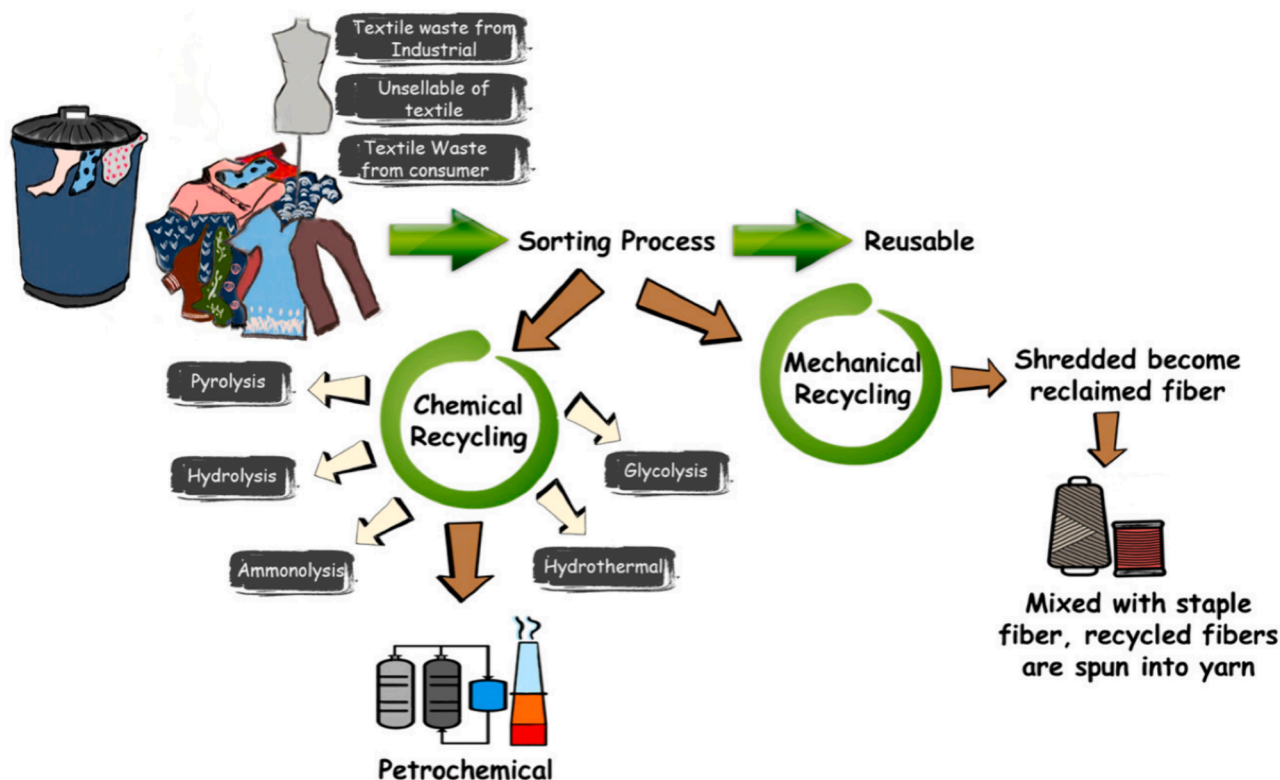


Fig. 3. Mechanisms of possible textile waste recycling routes [Reprinted from Damayanti et al., 2021 and is an open-access article (Copyright © 2021 by authors) distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license].

adsorption performance of the chars on Cr (VI), and the highest adsorptive capacity of char-FeCl₃ was 70.39 mg/g (Damayanti et al., 2021). In a study, Wang et al. (2020) investigated the catalytic pyrolysis of ultrastable zeolite. In another study, Yousef et al. (2021) new method for functionalisation of char obtained through pyrolysis of textile waste and its use.

Hydrothermal method

To decompose organic matter, the hydrothermal method uses chemical crystallisation at high temperatures and pressures. Hydrothermal method is a promising alternative for degrading carbon-polymer waste (Feng et al., 2021). This method uses water and can be divided into five categories such as hot water extraction, pressurised hot water extraction, treatment, hydrothermal carbonisation, and hydrothermal liquefaction based on the temperature range used (Elliott et al., 2015). The hydrothermal process requires a temperature near 280 °C. The hydrothermal method has a number of drawbacks, including long reaction times and high pressures (Li et al., 2017). In a study, researchers (Hongthong, 2021) studied about hydrothermal liquefaction for the conversion of nylon 6 net waste using macroalgae, *Fucus serratus*. A 50/50 mix of nylon 6 and *F. serratus* could produce up to 17 % bio-crude yield. Nylon 6 was completely degraded during the production of caprolactam, a molecule derived from nylon 6. Xu et al., (2021a) found that surfactants were used to catalyse hydrothermal carbonisation.

One step in the hydrothermal process involves splitting the polyester-cotton fibre waste mixture into smaller pieces, and another in dispersing that mixture with an organic acid catalyst in an aqueous phase to produce products (Hou et al., 2021). High pressure caused the reaction temperature to rise to 140 °C. This achieved a 99 % polyester fibre aggregate's recycling yield and an impressive 81 % of the cotton fibre fragments' recycling yield were also achieved. In a study, Qi et al. (2021) used FeCl₃ as a catalyst, lowering the hydrothermal carbonisation temperature to achieve highly efficient hydrophobicity.

Gasification

Gasification processes recycle textiles at higher temperature. Low or no oxygen is required for the reactions to take place. Gasification is more difficult than pyrolysis because it involves chemical reactions with the substance itself (Arafat and Jijakli, 2013). Gasification, like pyrolysis, produces gases with high concentrations of carbon and hydrogen. Although pyrolysis yields a higher percentage of gaseous waste, gasification yields a much higher gaseous fraction (Wu and Williams, 2010). When gasification is used, the increased pressure can lead to increased H₂ concentrations, calorific value, yield, and CO. In contrast, as operating temperatures rise, syngas production rises while char yield falls (Ahmad et al., 2016). It is possible to distinguish between various types of gasifiers by looking at the reactor bed and the flow pattern. The most common gasification methods use a fixed bed or fluidised bed gasifier. On the other hand, it is hard for coarser solid particles to separate because of the way the hydraulics are set up (Sridhar, 2012).

When using gasification, the goal is to produce synthesis gas, which includes compounds such as hydrogen, CO₂, and methane (CH₄). A mixture of ethylene and ash by-products can be formed. The steam gasification of mixed textile waste was studied by Vela et al. (2019) using a bench-scale fluidised bed reactor at 850 °C. Polyethylene glycol production yields polyester, cotton, and 50/50 polyester/cotton (Atakan et al., 2019). In another investigation, Vikrant et al. (2020) reported that textile wastes such as cotton, wool, and polyester had definite activation energies making them ideal candidates for this process.

Sustainable approaches to textile recycling

Biological method

Fibers that degrade in nature or through synthetic means are known as biodegradable because enzymes are able to break down their chemical bonds. Mechanical properties and degradability will determine whether or not a material is suitable for a given application (Azevedo and Reis, 2004). The degradation process may take two weeks to six months for the most biodegradable textile materials, including composites. Textile biodegradability depends primarily on crystallinity and orientation, hydrophilicity/hydrophobicity, soil conditions, and the types of microorganisms that live in them (Zambrano et al., 2020).

Fungi and bacteria cause degradation of cotton fabrics. Cotton fibres, primarily composed of cellulose, are well-known for their susceptibility to microbial degradation. *Bacillus*, *Fusarium*, *Clostridium*, *Myrothecium*, *Sporocytophaga* and *Memmoniella* are known microorganisms responsible for cellulose hydrolysis and oxidative degradation (Sanders et al., 2021). Fungi are efficient in the degradation of cellulosic fabrics (Zayed et al., 2022).

Laccase-mediator and protease systems have been used for polyamide 6.6 (nylon) degradation (Kabir and Koh, 2021). Bacteria, viz., *Flavobacterium* and *Pseudomonas* sp., can degrade nylon oligomers (Sanuth, 2012). Oxidase from lignolytic fungi have also been used to depolymerise polyamides. Keratinolytic enzymes play an important role in the decomposition of keratin-rich wastes from the wool textile industry (Wojnowska-Baryła et al., 2022). Keratinases are the proteases capable of degrading complex proteins (Jagadeesan et al., 2020), and keratin is used as a carbon, sulphur, nitrogen, and energy source by many bacterial, actinomycete, and fungal species (Bohacz, 2017). A wide range of environments from hot springs to Antarctic soils have been found to contain these microorganisms (Sahay et al., 2017). In a study, researchers studied about cotton based textile wastes and cellulose development through enzymatic saccharification (Hong et al., 2012).

Glycolysis

Degradation processes like glycolysis can be used to break down the large molecules of textiles into smaller ones. The short reaction time and reduced energy consumption are advantages of the glycolysis process. Polyethylene terephthalate (PET) fibres is one of the common textile materials (Park and Kim, 2014). A method to recycle PET fibre waste in an environmentally friendly way is required as they cause significant environmental harm (Awoyera et al., 2021). Glycolising PET waste for degradation is a difficult task because it requires the use of a catalyst (Hoang et al., 2019). The macromolecules of textile fibres are converted into monomeric forms. Guo et al. (2021) investigated the recycling of polyester textiles using Mg-Al as a catalyst, and found that using a double oxide catalyst with wet mixing, the BHET yield was over 82 %. Furthermore, the spinnability and mechanical properties of the re-polymerised recycled PET (r-PET) fibres were comparable to those of PET fibres. In a study, researchers (Damayanti and Wu, 2021) reported the possible ways of recycling PET.

Hommez and Goethals (1998) studied the depolymerisation of nylon 6 via the glycolysis process. The glycolysis of nylon 6,6 with ethylene glycol produced major products of the bis(hydroxyethyl) adipate and hydroxyethylester groups (Thiyagarajan et al., 2022). In a study, BHET achieved 100 % PET fibre conversion and 80 % monomer yield, respectively (Heiran et al., 2021). Dimers formed from the oligomers were unstable and had a low molecular weight due to the oligomer breakdown into dimers. The glycolysis of PET fibres to produce BHET with lithium hydroxide, acetic acid, sodium sulphate, and potassium sulphate, on the other hand, resulted in yields at different levels (Ghosal and Nayak, 2022).

Ammonolysis

Nylon 6,6 carpet waste is primarily depolymerised via the ammonolysis method. Lee et al. (2019) used ammonia or ammonium phosphate as a catalyst to study the reactions of nylon 66 and nylon 6 mixtures. In this process, the temperature and pressure were between 137 bar and 200 °C, respectively, and the main byproducts of nylon 6,6 depolymerisation were 5-cyanovaleramide, hexamethylenediamine, and adiponitrile. In another study, Chanda (2021) described also the depolymerisation mechanism for a nylon 66 and nylon 6 mixture.

Enzymatic hydrolysis

Enzymes produce excellent degradation results in textile recycling. Using enzymatic hydrolysis, Li et al., (2019a) were able to recover polyester and sugar compounds from textile wastes. The pretreatment was done at a temperature of 20 °C for 6 h with a NaOH/urea concentration of 7 percent/12 percent. With a cellulase dosage of 20 FPU/g, 98.3 % of the glucose was recovered. In hydrolysis reactions, cellulases act as catalysts. It is theoretically possible to reuse these enzymes indefinitely, as they are not affected by the equilibrium conditions of the reaction. The fact that these enzymes are organic compounds necessitates consideration of some decomposition processes that take place over time (Piribauer and Bartl, 2019). The degradation of these enzymes can alter the physicochemical properties of materials, which can also break down material structures. Certain enzymes are capable of initiating a wide range of enzymatic reactions. Certain parts of these materials may therefore be selectively broken down by enzymes. Commercial cellulase mixtures are typically composed of three types of enzymes, endoglucanases, exoglucanases, and beta-glucose depolymerases, all of which depolymerise the disaccharide cellobiose into monosaccharide units, e.g., glucose (Hu et al., 2018). Textile waste such as cotton, nylon, polyester, and silk can be recovered through the use of an enzymatic degradation process called fermentation.

Microbial engineering

Conventional methods can't decompose synthetic fibre blends due to low enzyme accessibility and a lack of efficient enzymes that can decompose synthetic materials quickly (Wojnowska-Baryła et al., 2022; Orlando et al., 2023). So, microbial engineering contributes to the breakdown of synthetic fibre (Buragohain et al., 2020). Focusing on materials with hydrolysable bonds helped in the enzyme-catalysed depolymerisation of textile polymers. Polyester made from terephthalic acid and ethylene glycol is known as polyethylene terephthalate (Jönsson et al., 2021). It was reported by researchers that a hydrolase from engineered *T. fusca* could degrade PET to some extent, and subsequent reports have demonstrated PET hydrolysis using, as well as PET degradation using other hydrolases (Yoshida et al., 2016). *I. sakaiensis* 201-F6 has recently been shown to degrade and grow on PET as a part of its carbon and energy source. PETASE (polyethylene terephthalate-degrading enzyme) and MHETASE [mono(2-hydroxyethyl) terephthalic acid hydrolase] are two enzymes discovered by microbial engineering that catalyse polymer hydrolysis. Several researchers discussed the role of genetically engineered PETase and MHETase in the degradation of PET that offers a sustainable strategy (Maity et al., 2021; Hachisuka et al., 2021). Site-directed mutagenesis experiments also allowed for a better understanding of the PET degradation mechanism and increased the enzyme's biocatalytic potential (Hajighasemi et al., 2016). It is well established for PA those proteases degrade keratinous proteins in wool (Harmsen and Bos, 2020).

Nylon-and nylon-oligomer hydrolases from *Agromyces* sp. have been reported to be useful in the processing of synthetic nylon. The *Flavobacterium* sp. KI72 have been shown to hydrolyse cyclic or linear nylon oligomers. When tested on Nylon-6 in low yields, manganese peroxidase from *B. adusta* showed activity through an oxidative mechanism

(Jönsson et al., 2021). White rot fungi IZU-154, *T. versicolor*, and *P. chrysosporium* were used to degrade Nylon 6,6. Laccases produced by *P. chrysosporium* can degrade polypropylene to some extent via oxidative mechanisms (Kale et al., 2015). Hydroquinone peroxidase from *A. beijerinckii* has shown some biodegradation activity for polystyrene. It has been found that gut microbes of mealworms help to depolymerise polystyrene foam. In a study, Gricajeva et al. (2022) reported the polyester plastic biodegradation by carboxyl ester hydrolase enzyme.

Textile waste valorisation to value added products

The problem of textile waste can be resolved with the help of existing innovative technologies and systems. As discussed in the following section, these technologies make it possible to acquire environmentally friendly alternatives for value added products (Fig. 4). There are different pretreatment approaches used for valorisation of wastes into useful bioproducts (Fallahi et al., 2021; Usmani et al., 2020, 2022). Biotechnological approaches have been reported as important in valorising textile waste (Stanescu et al., 2009). In a recent review, Mishra et al. (2022) discussed valorisation strategies for textile wastes and their uses in different sectors along with challenges in different processes.

Bioethanol

Pretreatment before bioethanol production is a critical step that must be done correctly. By altering the porosity and crystallinity of cellulose feedstocks, pre-treatment increases the surface area and accessibility for enzymes. Chemical (alkali) and physico-chemical pretreatment as well as combinations of the two have been investigated for this purpose (Devi et al., 2021). In a study, Wojnowska-Baryła et al. (2022) achieved the highest ethanol production of 32.3 g/L (corresponding to a yield of 47.6 percent) using a sequential combination of organo-solvent and hydrothermal pretreatment of textiles. Six hours of pre-hydrolysis was done to form the highest glucose concentration for fermentation. Alkaline pretreatments of textile waste can improve ethanol production. In another study, Nemerow (2010) used different alkali solutions to pretreat polyester-cotton textiles. After that, a 72 h enzymatic hydrolysis process at 45 °C and pH 4.8 was performed on both treated and untreated textile waste.

Fermentation processes employing enzymes can recover valuable molecules and bioproducts from textile wastes. Prior to studying ethanol production via simultaneous saccharification and fermentation (using *S. cerevisiae*), researchers looked at the pretreatment conditions that produced the highest hydrolysis yields. According to the study's findings (Wojnowska-Baryła et al., 2022), all pretreatments were found to improve ethanol yields. A pretreatment of NaOH/urea at 20 °C resulted in the highest ethanol yield of 70 %. Textiles that had not been treated had a yield of 36 %. Cotton waste from textile mills was broken down using acid and alkali so that enzymes produced by *Fusarium* species could break down the sugars. More sugar was released from the acid pretreated substrates than from the alkali pre-treated substrates. *S. cerevisiae* was used to saccharify and ferment the sugars simultaneously. Batch fermentation produced 11.8 mg/mL of alcohol (Day, 2009).

Cellulose extracted from textile waste can be used as a source of ethanol (Rajendran et al., 2015). The cellulosic components of waste jeans were subjected to hydrolysis in alkaline pretreatment. Using commercial cellulase and glucosidase as the starting materials allowed *S. cerevisiae* to ferment the mixture and produce ethanol as the end product. Cotton and textile waste yielded comparable results. Cotton and cotton-based waste have ethanol yields of 69.4 percent and 59.5 percent, respectively, making this process interesting (Jeihanipour, 2011). In another study, Varadharajan (2014) revealed that cellulosic cotton waste could be raw material for bioethanol production through enzymatic hydrolysis followed by fermentation process via *S. cerevisiae*. He observed that acid pre-treated substrate elevated the enzyme action

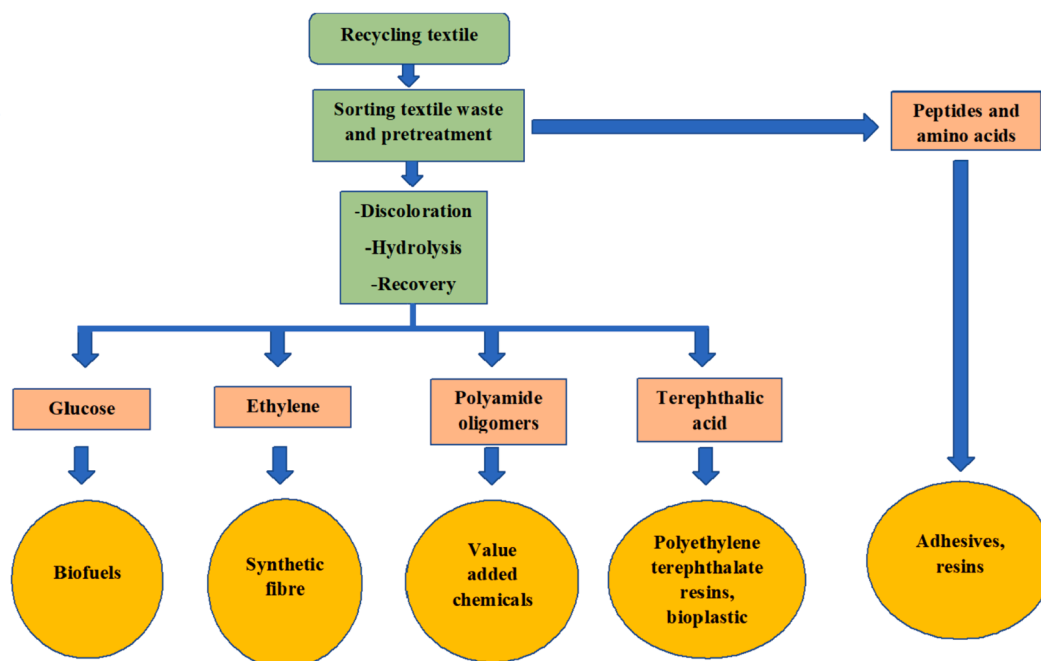


Fig. 4. Textile waste utilisation for value-added products.

that releases increased amount of sugar (90 mg/mL) compared to the alkali pre-treated substrates which results in 63 mg/mL of sugar release. Similarly, Nikolić et al. (2017) highlighted the potential of pre-treated cotton fabrics and waste textile products as a promising feedstock for renewable bioethanol production. They reported that the pre-treatment led to an increase in the glucose yield during enzymatic hydrolysis which results in bioethanol production through fermentation process by *S. cerevisiae* var. *ellipsoideus*. These findings revealed that textile waste materials can be a good renewable feedstock for producing biofuel like ethanol.

Biogas

Textile molecular structure significantly impacts gas production efficiency when it comes to cotton, polyester, viscose, and polyester biogas produced without pretreatment or milling (Pensupa et al., 2017). Textile pretreatment significantly impacted biogas production in a semi-continuous process. The initial biogas production rate in a two-stage batch process was higher. Because of the use of two stages of anaerobic digestion and the use of N-methylmorpholine-N-oxide (NMMO) on denim textiles, it was possible to achieve a higher organic loading rate (OLR) while keeping the hydraulic retention time (HRT) shorter (Jei-hanipour et al., 2013). Anaerobic digestion of cotton textiles has been shown to be affected by the substrate-to-inoculum ratio (SIR), in addition to the effect of chemical pretreatment on this process. It was found that different SIRs had different effects on wastewater from microcrystalline cellulose production when tested (Juanga-Labayen et al., 2021). The highest methane production rate and a kinetic constant of 0.24 d⁻¹ were obtained with a SIR of 0.5, resulting in a total yield of 8 days after incubation. Due to the buildup of acids, a SIR of 2.0 resulted in a process shutdown. The anaerobic digestion of untreated cotton textile waste was studied by Akinyele (2014). The cotton fabric was treated with 0.5 M Na₂CO₃ for three hours at 150 °C. Untreated cotton textile waste with a SIR of 0.5 produced the most methane (366.76 mL/g VS), indicating an anaerobic degradability of 89.67 percent. Treated cotton textile waste broke down anaerobically by about 75 %, which was equal to about 306 mL of CH₄/g VS (Juanga-Labayen and Yuan, 2021).

Bioplastics

Cotton waste from the textile industry can be used to produce biopolymer composites that are environmentally friendly and cost-effective. Before being formed into pieces by injection moulding, the as-received linter of recycled cotton was first chopped into short fibres known as recycled cotton fibres (RCFs) (Zaghloul et al., 2021). When RCF was added to bio-PET, it increased rigidity and hardness in the 1–10 wt% range. However, the ductility and toughness were significantly reduced, especially at the highest fibre contents. RCF was an effective nucleating agent for bio-PET crystallisation and also increased thermal resistance in the system. Interestingly, rigid food packaging and other applications could benefit from the biopolymer composite pieces, which can also help to improve biorefinery system design and the value of textile wastes (Eslami-Farsani et al., 2014). Analytical techniques were used to evaluate the films' physical, chemical, and mechanical properties. Creating value-added bioproducts from low-quality cotton fibres are also important (Sadeghifar et al., 2017). A circular textile recycling initiative (Dini, 2021) might be able to make cellulase from textile waste by using textile waste as a carbon source for SmF with filamentous fungus. Textile waste can be hydrolysed to recover cellulase, which can then be used to extract fermentable sugar and the remaining polyester fibre.

Renewable textile polymers, PLA fibers and synthetic fibres

Fibre recycling is a mechanical process that keeps the fibres intact while the fabric is disassembled and reused. These techniques are now used to process all types of textiles. Textile fibres are typically recycled mechanically through shredding (Harmsen et al., 2021). Human intervention is required before textiles can be shredded into smaller pieces, such as removing zippers and buttons. To make new yarn, slivers of the shredded textile are torn apart. Recycled yarn often needs to be blended with virgin fibres in order to meet industry standards for strength and quality (Ribul et al., 2021). On an industrial scale, a mechanical recycling process for cotton fibres has been developed (Schuch, 2016). Recycled and virgin fibres are combined to produce new yarn as part of the cotton industry's semi-closed-loop recycling process. Wool's closed-loop process eliminates the need for virgin fibres in recycling. To

describe fiber-to-fiber wool recycling in the Prato area, Italian producers use the term “regenerated wool.” Down-cycling is used in an open-loop process because the quality of the fibre decreases if it is recycled again. In an open-loop process, polyester and acrylic waste have been turned into a non-woven fabric or insulation (Mikolajczak, 2015).

Thermomechanical recycling

Synthetic fibres must be melted in order to be respun into new fibres or reshaped into new shapes via thermomechanical recycling. Elastane fibres can be broken if they are not re-spun into solid shapes such as bars before they are put through the thermomechanical recycling process, which is used for pure synthetic fibres such as thermoplastic polyurethane, nylon 6, polyamide 6 and elastane (Harmsen et al., 2021). Despite this process being possible and appearing a good way to recycle polyester, it hasn't been used on a large scale yet because it's expensive to get rid of contaminants, with the exception of a few commercial examples like Teijin fibers (Palm et al., 2014).

Monomer recycling. To obtain monomers in their original form, the polymer chain must be disassembled in the process of monomer recycling. Thereafter, the monomers can be polymerised into new virgin polymers. This method has been used to create new fibres from a variety of textile waste. Some examples include Nylon 6, recycled from carpets and fishing nets (MacArthur, 2017). Open-loop recycling of PET plastic into textile fibres has been well-documented using a variety of chemical techniques. It has been suggested that sodium formaldehyde sulfoxylate is a solution to polyester dyes that hinder closed-loop recycling (Fei et al., 2020). Mg–Al double oxide pellets have recently been used as a catalyst in the recycling of post-industrial PET textile waste that is melt-spun into fibres (Šišková et al., 2021). The monomer recycling process has been tried for textile materials made of cellulose and wool, but so far it hasn't led to textiles.

Polymer recycling

Mechanical processes such as shredding separate the fibres, which are then dissolved chemically using specific, often hazardous solvents. Fibres are spun and regenerated while the polymer is preserved (Thio-unn and Smith, 2020). Chemicals such as bleach are used to remove dyes during these processes. In recent years, ionic liquid solvents have been used to reintroduce dyed post-consumer textile waste into a new life-cycle, and this significantly reduces its environmental impact (Al-Salem et al., 2009). Viscose and lyocell processes for making man-made fibres from wood pulp were the precursors to chemical polymer recycling of post-consumer cotton or cellulosic byproducts. Cotton waste clothing with and without wood pulp have been developed using the lyocell process, as have fibres made from pre- and post-consumer waste combined with wood pulp (Nguyen, 2020). Ionic liquids (ILs) can dissolve cellulose instead of NMMO. An IL is typically a salt that has a melting point lower than 100 °C (Hermanutz et al., 2019). To regenerate cellulose-based textile fibres, ILs have been used to dissolve post-consumer cotton waste textiles (Schuch, 2016). A more environmentally friendly alternative to N-methylmorpholine-N-oxide (NMMO), derived from the Lyocell process, is an ionic liquid, which can dissolve cellulose at room temperature without degrading either the solvent or the cellulose. Companies like Circulose and Nucycle, are currently commercialising the chemical polymer recycling of textiles made of cellulose (Kim et al., 2010). Cellulosic by-product recycling has also benefited from polymer recycling. Orange peel, a byproduct of the juice industry, is used as a cellulose source by Italian company Orange Fiber to make textile fibres for yarns and fabrics. Another means to develop value while also reducing the negative effects on the environment is through the valuing of industrial waste yarn (Kour et al., 2019).

Other value-added chemicals

Succinic acid

Succinic acid can be made from cotton textile waste hydrolysate in a fibrous bed bioreactor with a high glucose content (Li et al., 2019b). Producing succinic acid from inexpensive starting materials such as tetrahydrofuran, 1,4-butanediol, succinate polymers, and adipic acid is possible. Succinic acid serves as a chemical building block for a variety of high-value-added chemicals used in the detergent/surfactant, food, and pharmaceutical industries (Molino et al., 2020). Microbial fermentation is the primary method used in bio-based production processes (Nghiem et al., 2017).

Cellulase

Textile waste can also be utilised as a low-cost feedstock for producing cellulase enzymes (Hu et al., 2018). In a study, Wang et al. (2018) reported cellulase production through submerged fungal fermentation using textile waste as a feedstock. They found after growing on textile blending cotton/polyester 40/60 as substrate, *Trichoderma reesei* was observed with the highest cellulase activity (18.75 FPU/g). These findings indicate a high potential for processing textile waste for the recovery value-added products.

Building construction material from textile waste

Textile waste can be used to produce raw materials for common construction applications, such as thermal insulation, noise insulation, and concrete fillers. Resins are also used in construction as adhesives and coatings. Soil reinforcement using fibrous carpet waste showed that carpet waste-derived fibrous inclusions improved the shear strength of silty sands (Hu et al., 2018). Textile reinforced concrete (TRC) is a composite concrete that uses textiles as reinforcement. It is used in a wide range of situations, such as repair and rehabilitation work, precast construction, and structural reinforcement of existing structures. By repurposing textile industry waste, the construction industry is promoting the use of sustainable building materials (Scheerer et al., 2017). Because it combines fine-grained concrete and multi-axially oriented textiles, the material possesses a strong, thin, and corrosion-resistant structure with excellent ductility, superior strength, and no magnetic disturbances (as found in steel reinforced concrete).

Thick ropes can be made from textile waste, with applications such as to protect slopes from sliding and erosion. Recycled scraps of wool and nonwoven fibres were repurposed into ropes (Briga-Sa et al., 2013). The findings confirmed the effectiveness of such technology in protecting steep slopes.

Advantages and challenges in textile waste management

The many different technologies used to break down textiles and turn them into usable products each have their own set of benefits and challenges, outlined in Table 1.

Future directions and socio-economic conditions

The future of recycling textiles depends on the creation of new ways to reuse, reduce, and recover value-added products. Recycling textiles could form a large part of a sustainable bio and circular economy. To solve the textile waste problems, there is a need for innovative and cross-disciplinary research that focuses on different biotechnologically advanced strategies. Also, a strong policy environment is required to restrict improper and direct discharge of textile wastes and products into the environment. Governments, the scientific community, and businesses should all take steps to work together on the possible opportunities for recycling and waste utilization in this area. With the help of financial aid from the government, significant business opportunities exist in recycling to make products with added value. Understanding

Table 1
Recyclable textile wastes and their technological challenges.

S. No.	Technologies to recycle textile wastes	Advantages	Challenges	References
1	Pyrolysis	- It's a Simple process. - Applied on broad range of materials. - Does not necessitate prior textile waste treatment.	- High-temperature chemical reaction process. - Excessive use of electrical power.	Yousef et al., 2019; Zhou et al., 2022
2	Hydrothermal	- Low ash and oxygen. - Less temperature compared to pyrolysis.	- Long reaction time - Decreased purification, as well as heterogeneity.	Matayeva and Biller, 2022; Xu et al., 2021b
3	Gasification	It can be used with potentially combined textile waste	- High consumption of energy - High degree of heat is required.	Rittfors, 2020; Vela et al., 2019
4	Novel approaches			
4.1	Biological	- This method has a high probability of removing dyes from clothing. - An eco-friendly solution. - Low energy consumption	Toxic intermediates are produced at a high cost and have a long retention time compared to other methods.	Hussain et al., 2021; Saini, 2017.
4.2	Glycolysis		- Low selectivity. - Sluggish processing in the absence of a catalyst	López-Fonseca et al., 2010; Li et al., 2022
4.3	Ammonolysis	Textile waste can be used to make byproducts like carpet and upholstery.	- A blend of primary and secondary amines is produced. - Toxic solvents were used (ammonia) Costs rise due to the necessity for high pressure and temperature.	George and Kurian, 2014; Patti et al., 2020
4.4	Enzymatic hydrolysis	- A new microorganism boosts biopolymer production and conversion. It can be spun into new waste fibres. - Low energy demand. - Makes use of eco-friendly chemicals and solvents	- Enzymatic hydrolysis can only be used to recycle certain materials, such as rayon, hemp, cotton. - Requires a lot of water.	Li et al., 2019a,b; Piribauer et al., 2021
4.5	Microbial bioengineering	- More efficient to use genetically modified microorganisms. - Reduce the demand for energy.	Absence of effective enzymes for the rapid decomposition of synthetic man-made materials	Schiro et al., 2021; Pensupa et al., 2017

why people discard clothing and devising alternate methods for valorising unused clothing within our current systems to prevent landfilling and additional resource extraction will lessen the fashion industry's environmental consequences in future (Degenstein et al., 2021).

Conclusions

Recycling of textiles and their reuse is necessary for conserving natural resources and lowering environmental toxicity of the fashion and textile industries. Recycling also reduces the amount of energy and chemicals needed to make usable textiles, while reducing pollution generated during manufacturing. The degradation of fibres and the reduction of environmental pollution can be achieved through various physicochemical and biological valorisation technologies.

Recycling fibres into value-creating products such as biofuel, bioplastic, various chemicals, carpets, enzymes, acids, synthetic fibres, and resin is now possible. These methods can achieve sustainable growth in the textile and clothing industries, and they can be further improved through research into increasing efficiency and effectiveness and through scaling up of existing and improved techniques.

CRedit authorship contribution statement

Manikant Tripathi: Writing – original draft, Writing – review & editing. **Minaxi Sharma:** Writing – original draft, Writing – review & editing. **Saroj Bala:** Writing – review & editing. **Vijay Kumar Thakur:** Writing – review & editing. **Anoop Singh:** Writing – review & editing. **Kavya Dashora:** Writing – review & editing. **Phil Hart:** Writing – review & editing. **Vijai Kumar Gupta:** Conceptualization, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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